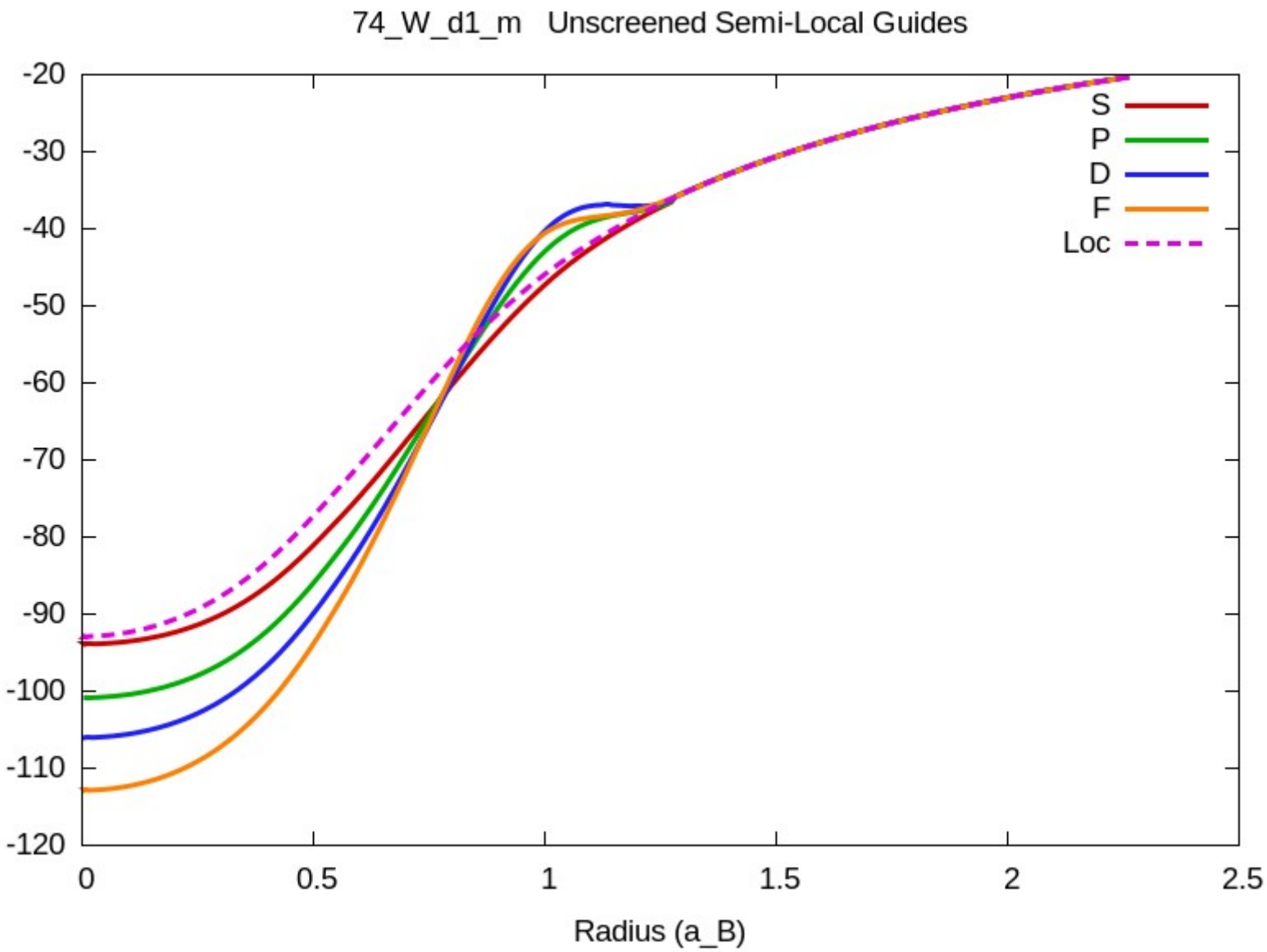
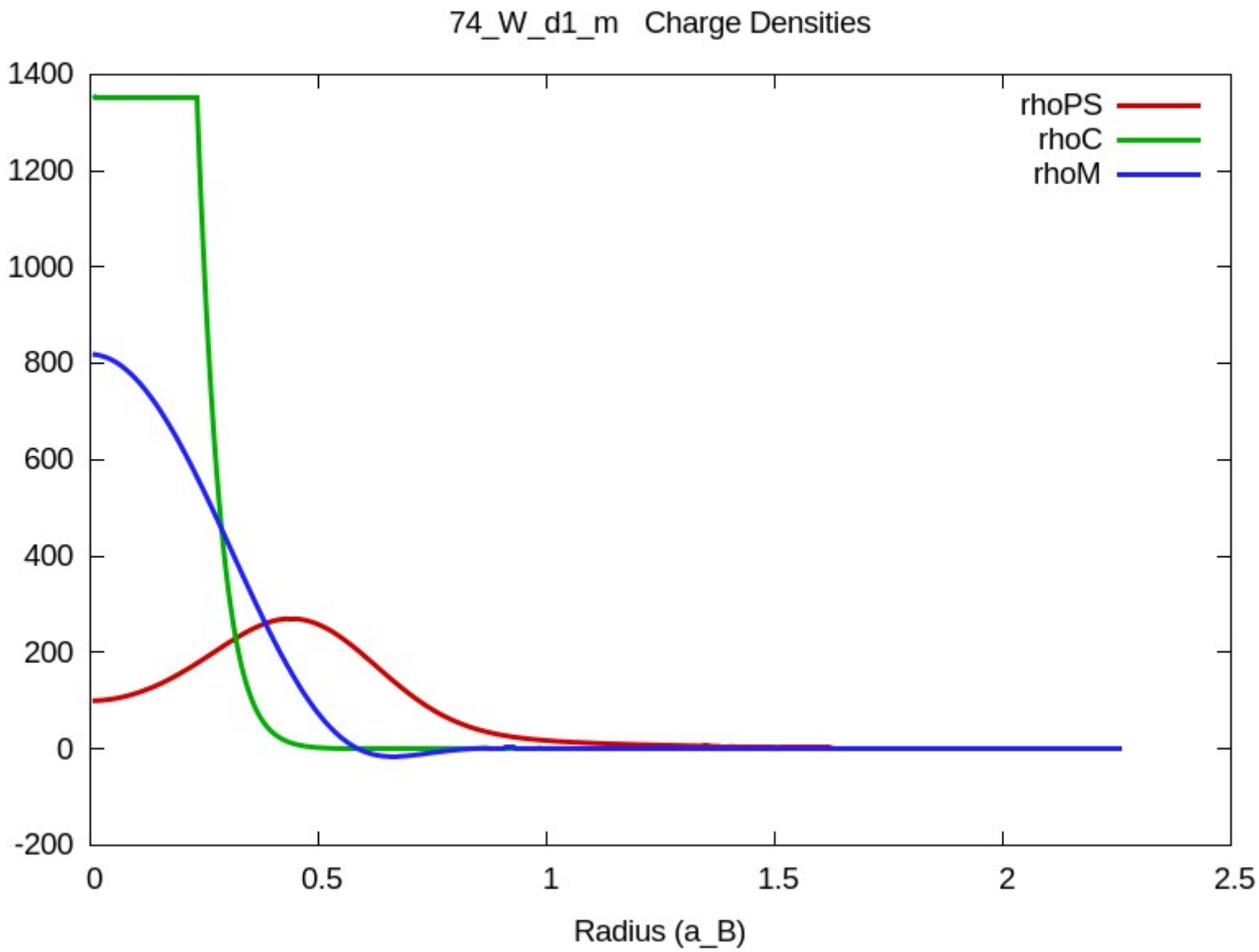
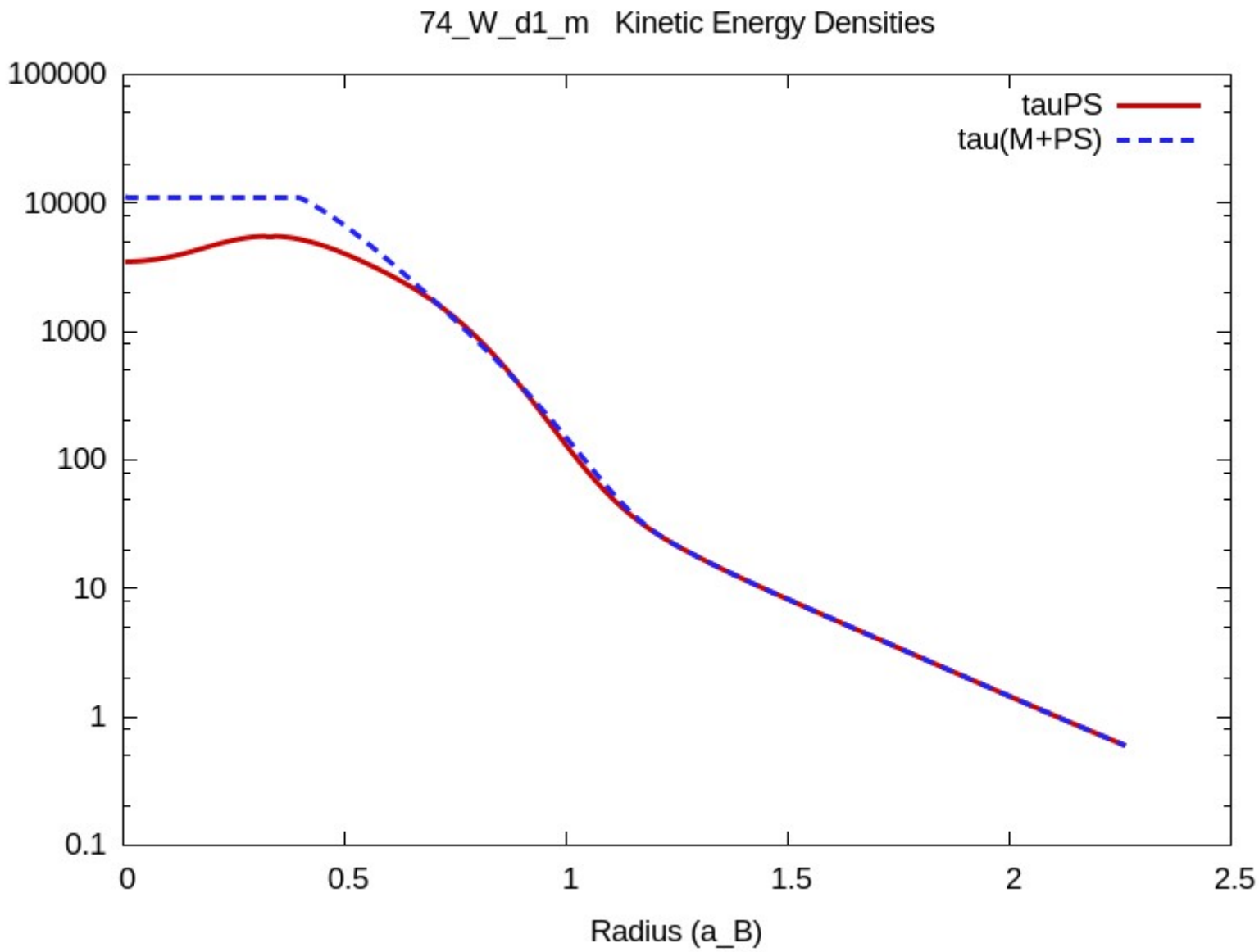
74_W_d1_m.out

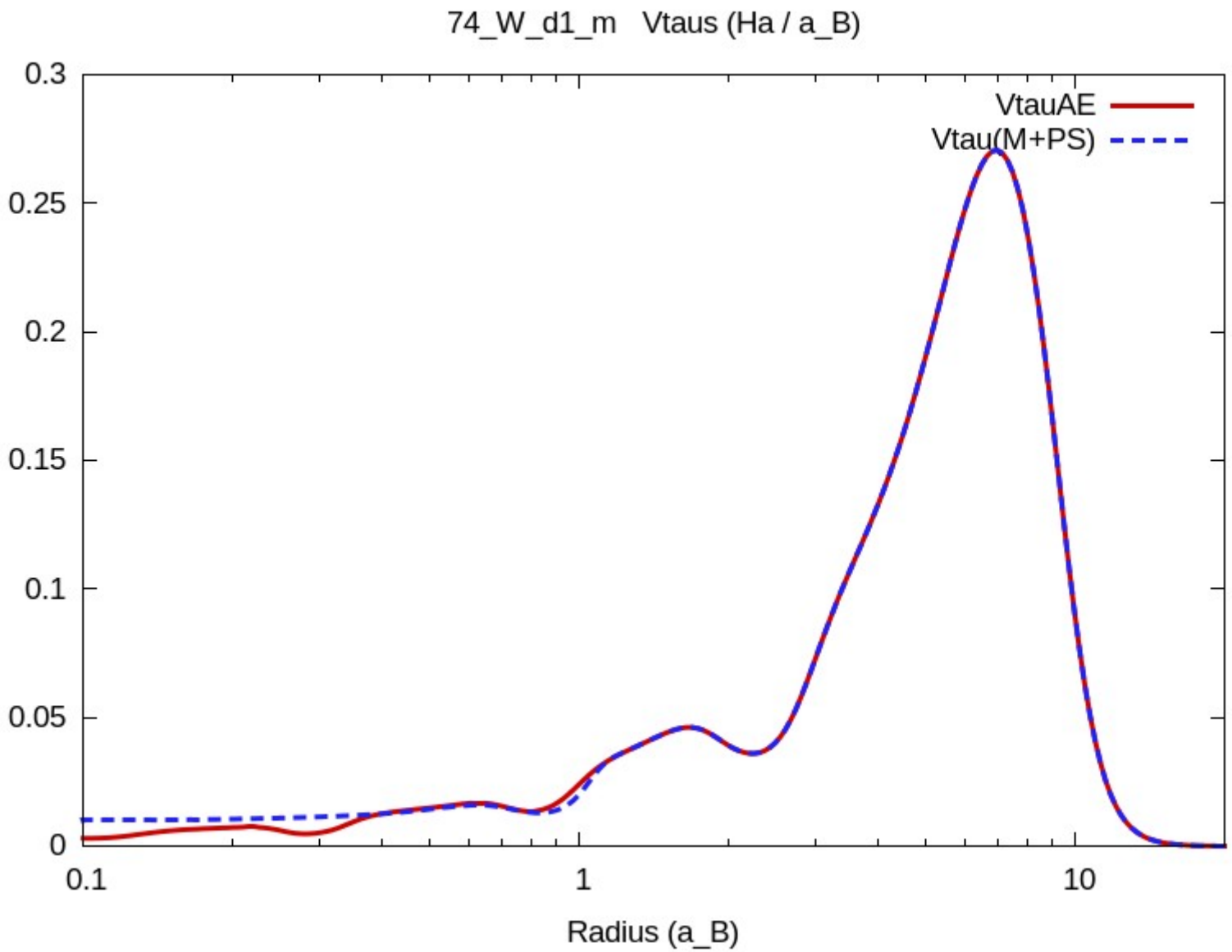
Diagnostic tests using Vanderbilt-Kleinman-Bylander pseudopotentials
1 or more projectors used as specified by nproj input data

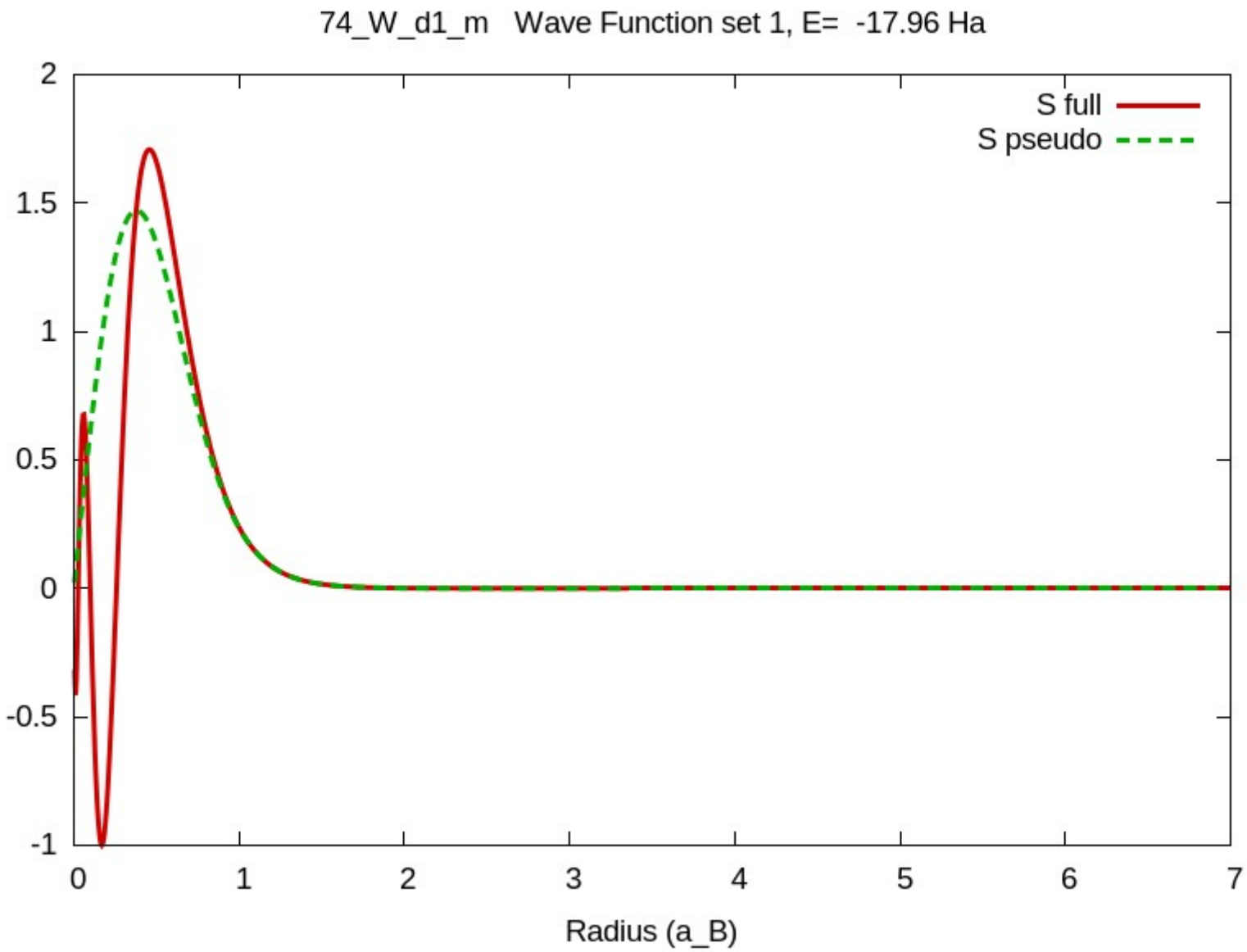
l	rcore	rmatch	e in	delta e	norm test	slope test
0	1.2735667	1.2927851	-17.9574545	0.0000000	1.0000000	1.0000000
0	1.2735667	1.2927851	-2.4611612	0.0000000	1.0000000	1.0000000
0	1.2735667	1.292785 1	-0.1803995	0.0000000	1.0000000	1.0000000
1	1.2735667	1.2927851	-14.7901033	0.0000000	1.0000002	1.0000002
1	1.2735667	1.2927851	-1.5450436	0.0000000	1.0000000	1.0000000
1	1.2735667	1.2927851	1.7549564	0.0000000	1.0000000	1.0000000
2	1.2735667	1.2927851	-9.0353391	0.0000000	1.0000000	1.0000000
2	1.2735667	1.2927851	-0.2156528	0.0000000	1.0000000	1.0000000
2	1.2735667	1.2927851	2.8143472	0.0000000	1.0000000	1.0000000
3	1.2735667	1.2927851	-1.5639872	0.0000000	1.0000000	1.0000000
3	1.2735667	1.2927851	4.7360128	0.0000000	1.0000000	1.0000000
3	1.2735667	1.2927851	11.0360128	0.0000000	1.0000000	1.0000000

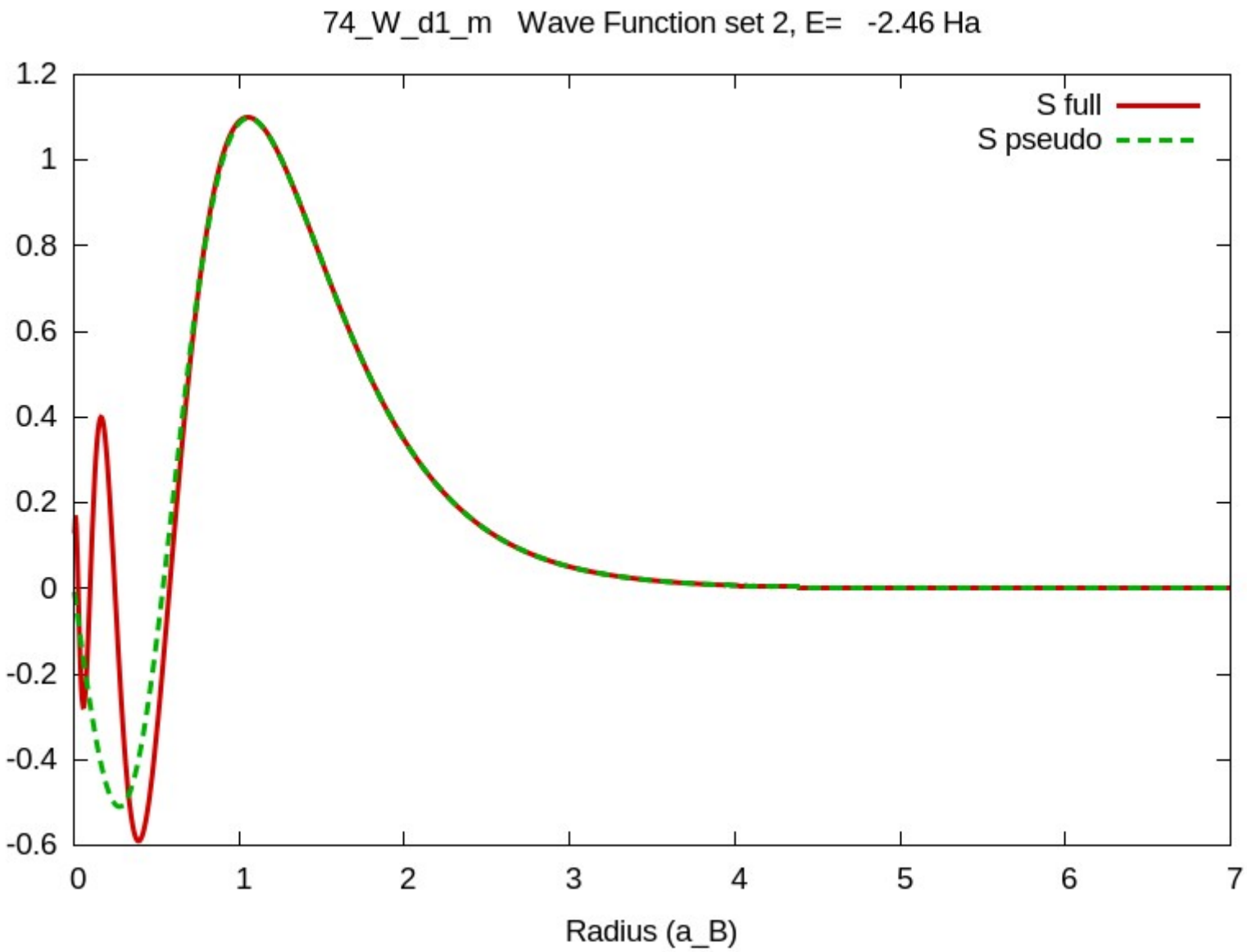

74_W_d1_m Unscreened Semi-Local Guides
-20
-30
-40
-50
-60
-70
-80
-90
-100
-110
-120
0
0.5
1
1.5
2
2.5
Radius (a_B)
S
P
D
F
Loc

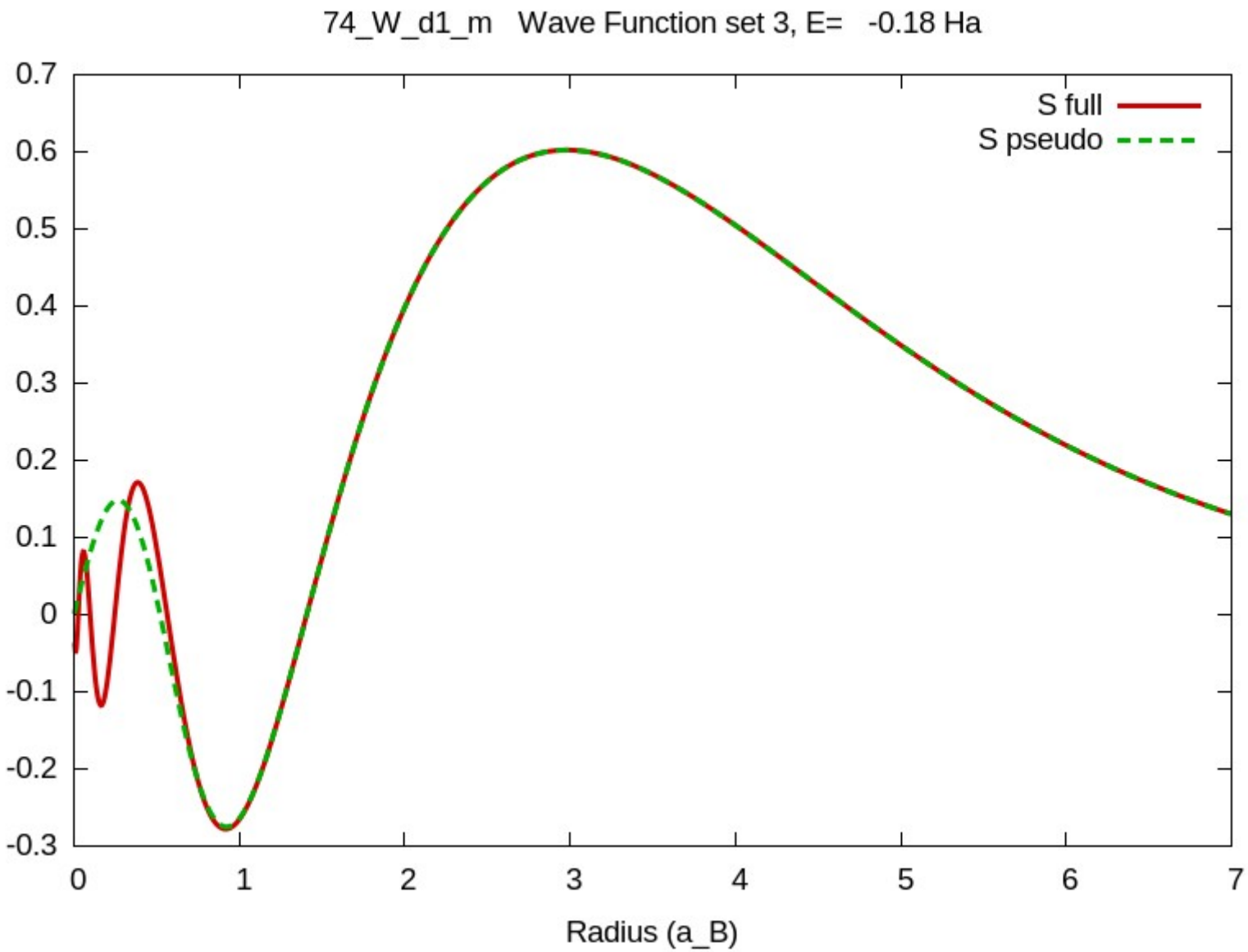

74_W_d1_m Charge Densities
rhoPS
rhoC
rhoM
1400
1200
1000
800
600
400
200
0
-200
0
0.5
1
1.5
2
2.5
Radius (a_B)

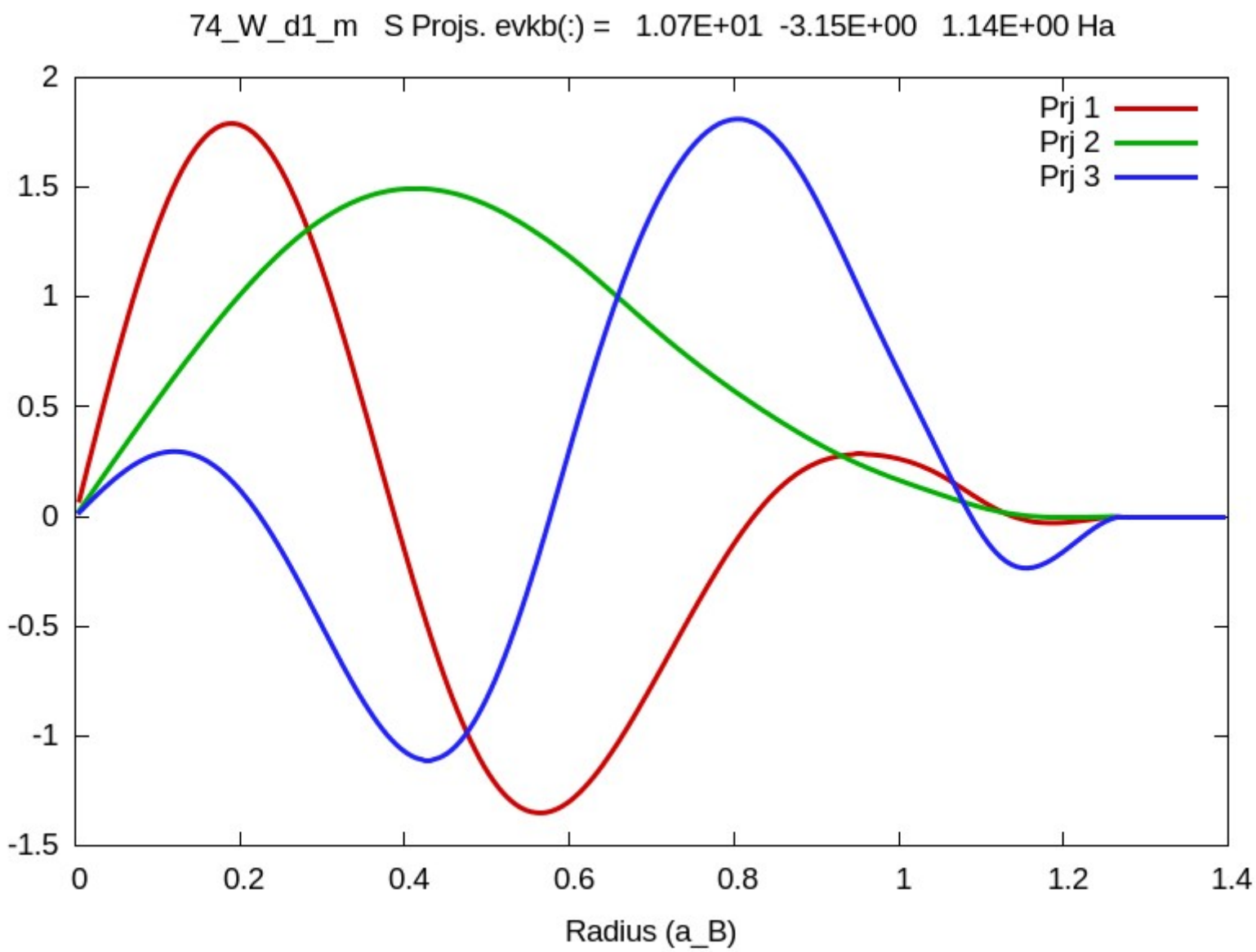

74_W_d1_m Kinetic Energy Densities
tauPS
tau(M+PS)
100000
10000
1000
100
10
1
0.1
0
0.5
1
1.5
2
2.5
Radius (a_B)

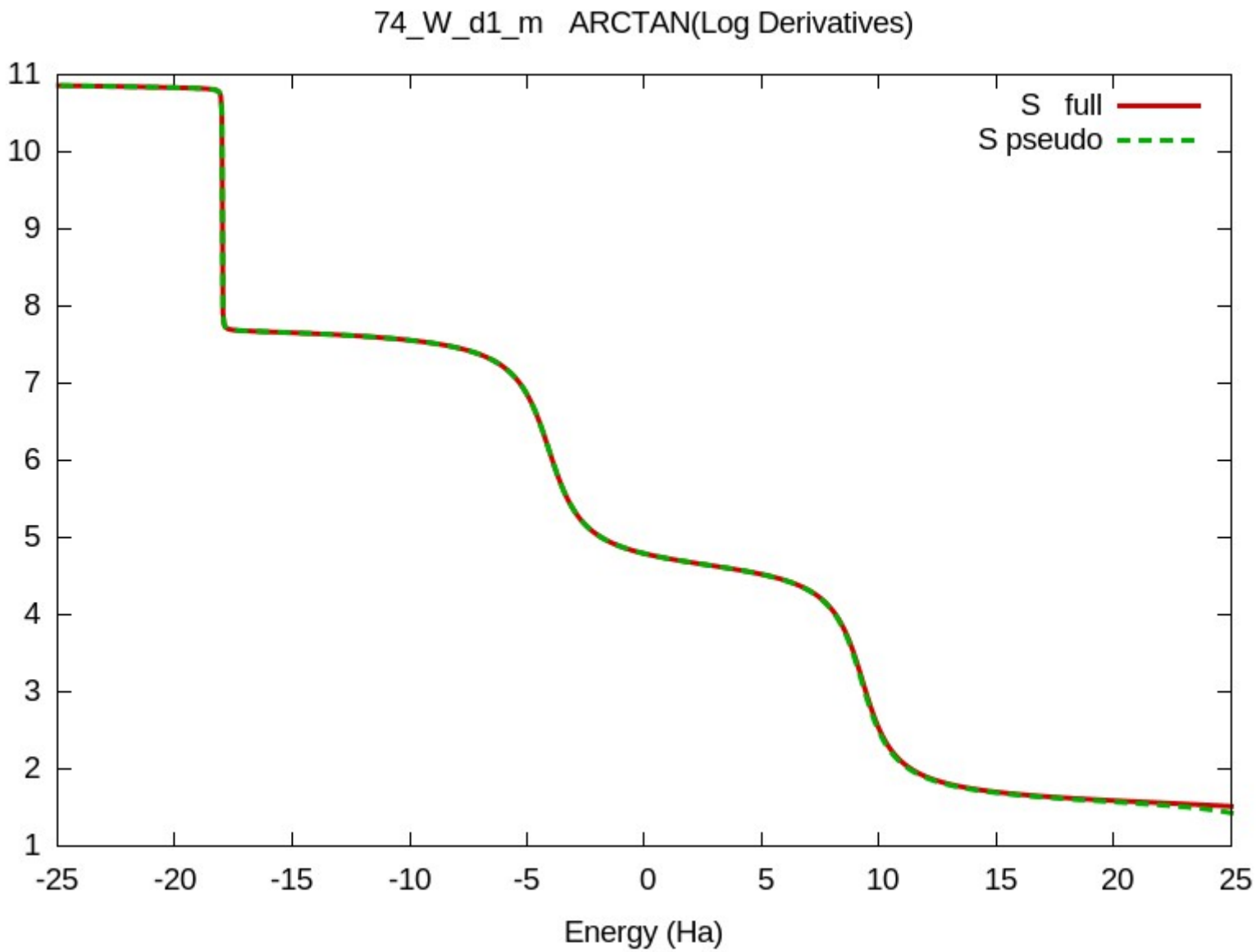

74_W_d1_m Vtaus (Ha / a_B)
VtauAE
Vtau(M+PS)
0.3
0.25
0.2
0.15
0.1
0.05
0
0.1
1
10
Radius (a_B)

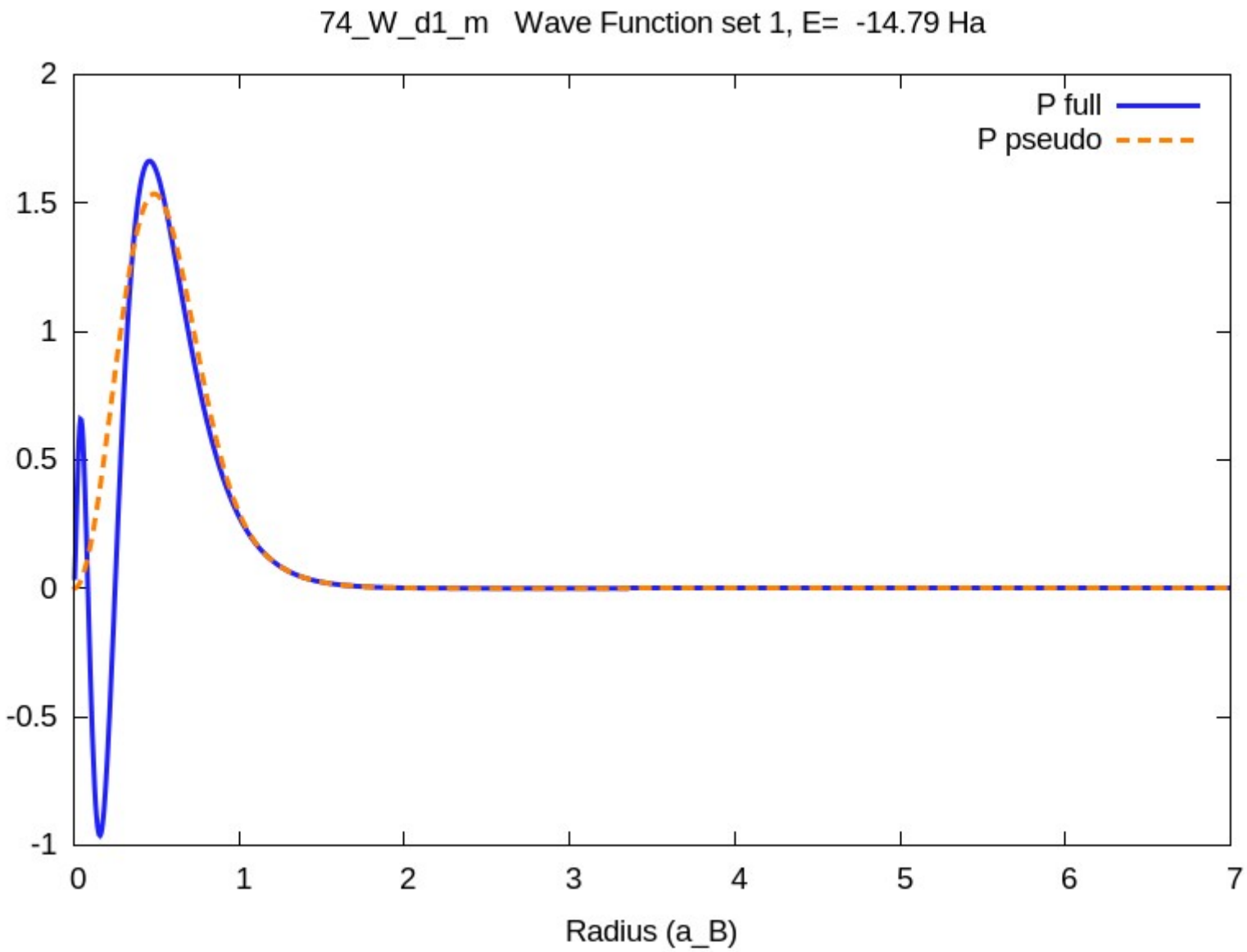

74_W_d1_m Wave Function set 1, E= -17.96 Ha
S full
S pseudo
2
1.5
1
0.5
0
-0.5
-1
0
1
2
3
4
5
6
7
Radius (a_B)

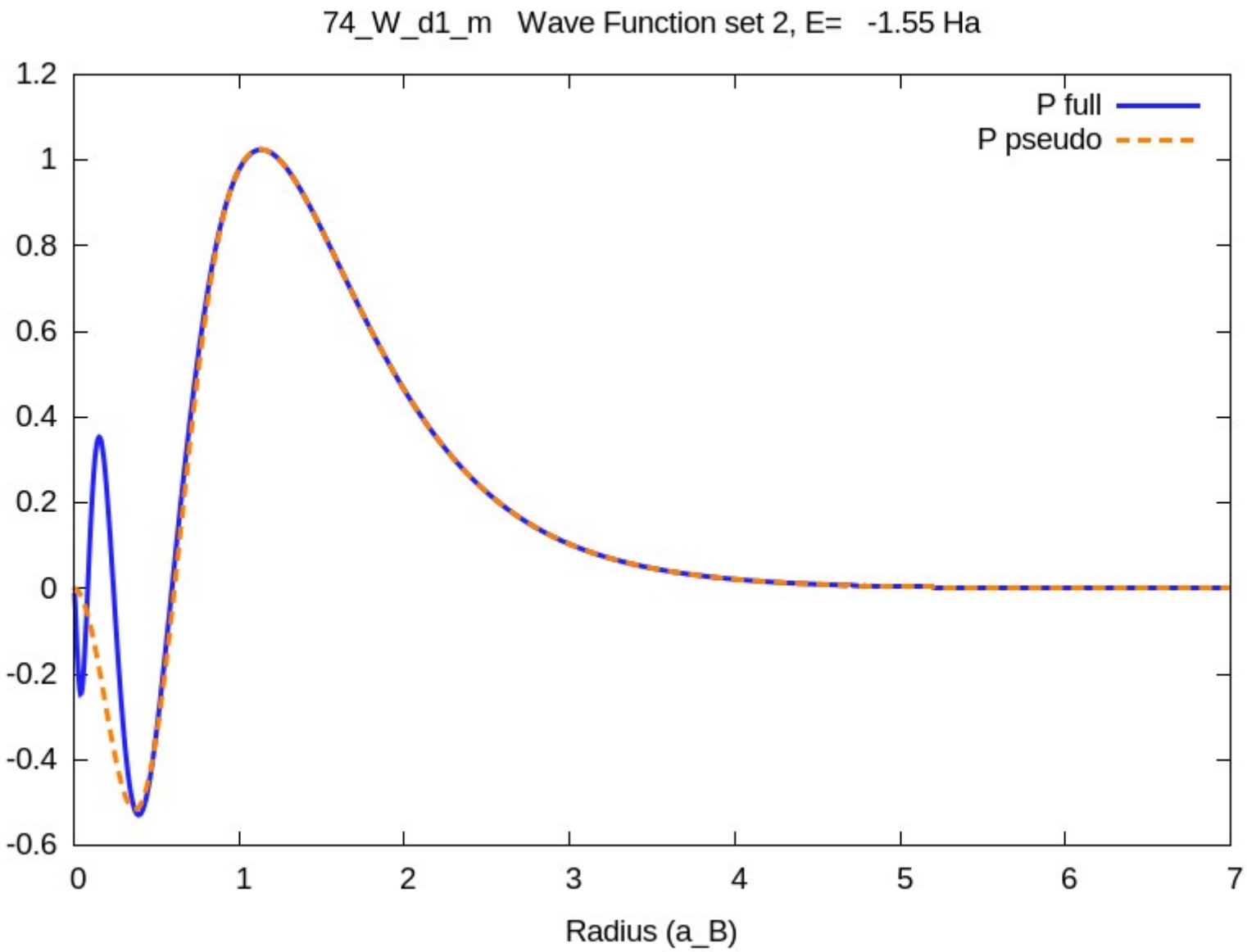

74_W_d1_m Wave Function set 2, E= -2.46 Ha
S full
S pseudo
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
1
2
3
4
5
6
7
Radius (a_B)

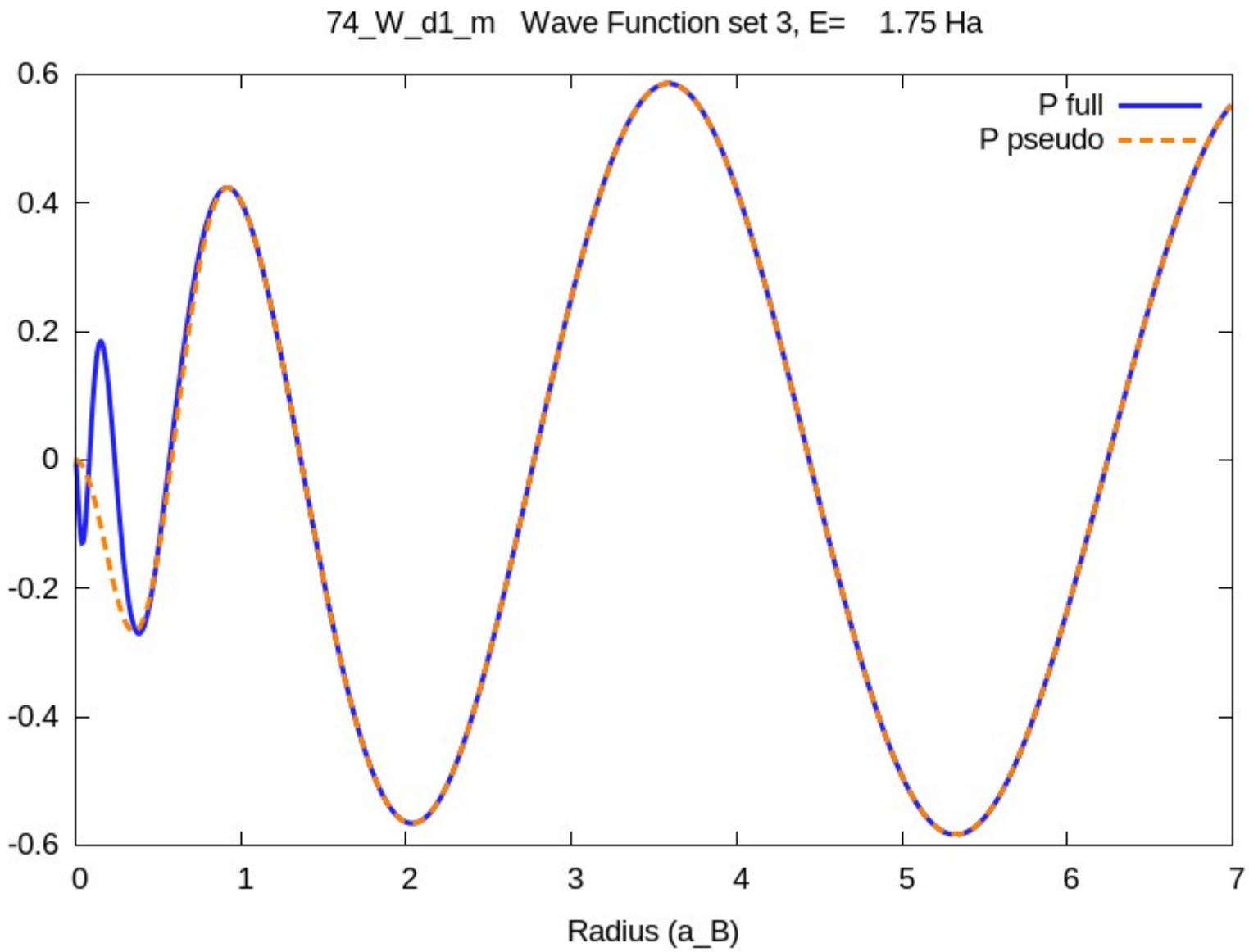

74_W_d1_m Wave Function set 3, E= -0.18 Ha
S full
S pseudo
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
0
1
2
3
4
5
6
7
Radius (a_B)

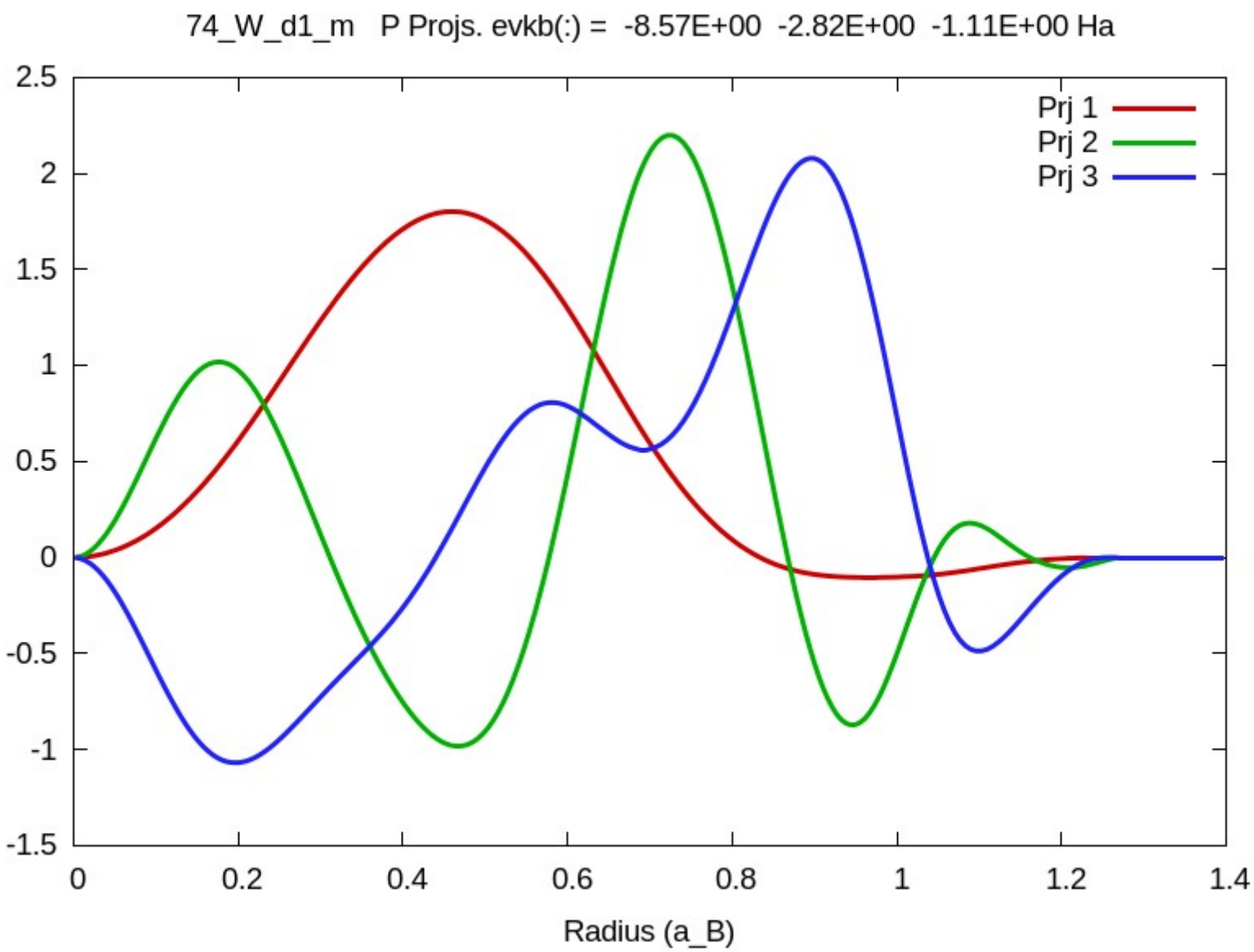

74_W_d1_m S Projs. evkb(:) = 1.07E+01 -3.15E+00 1.14E+00 Ha
Prj 1
Prj 2
Prj 3
2
1.5
1
0.5
0
-0.5
-1
-1.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
Radius (a_B)

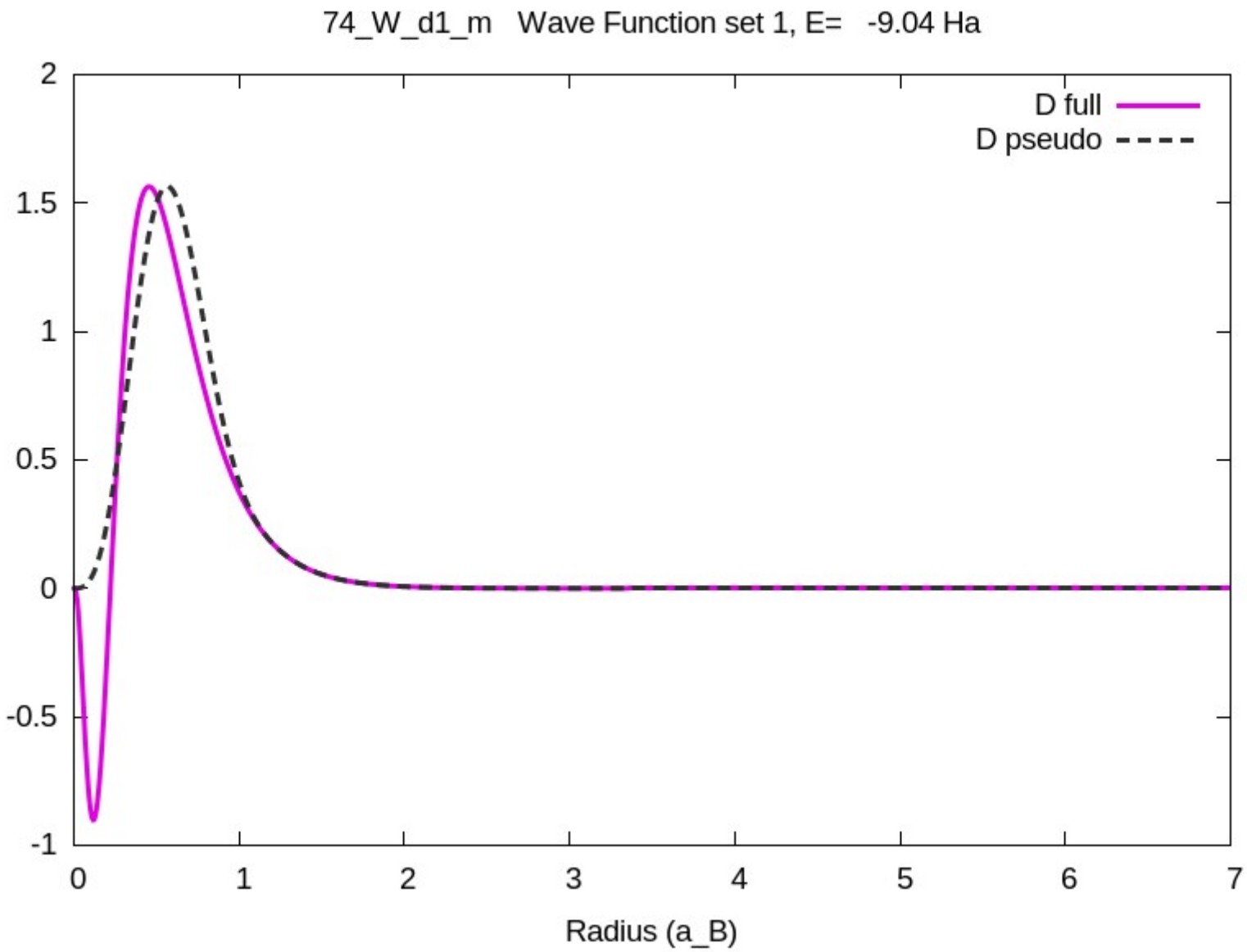

74_W_d1_m ARCTAN(Log Derivatives)
S full
S pseudo
11
10
9
8
7
6
5
4
3
2
1
-25
-20
-15
-10
-5
0
5
10
15
20
25
Energy (Ha)

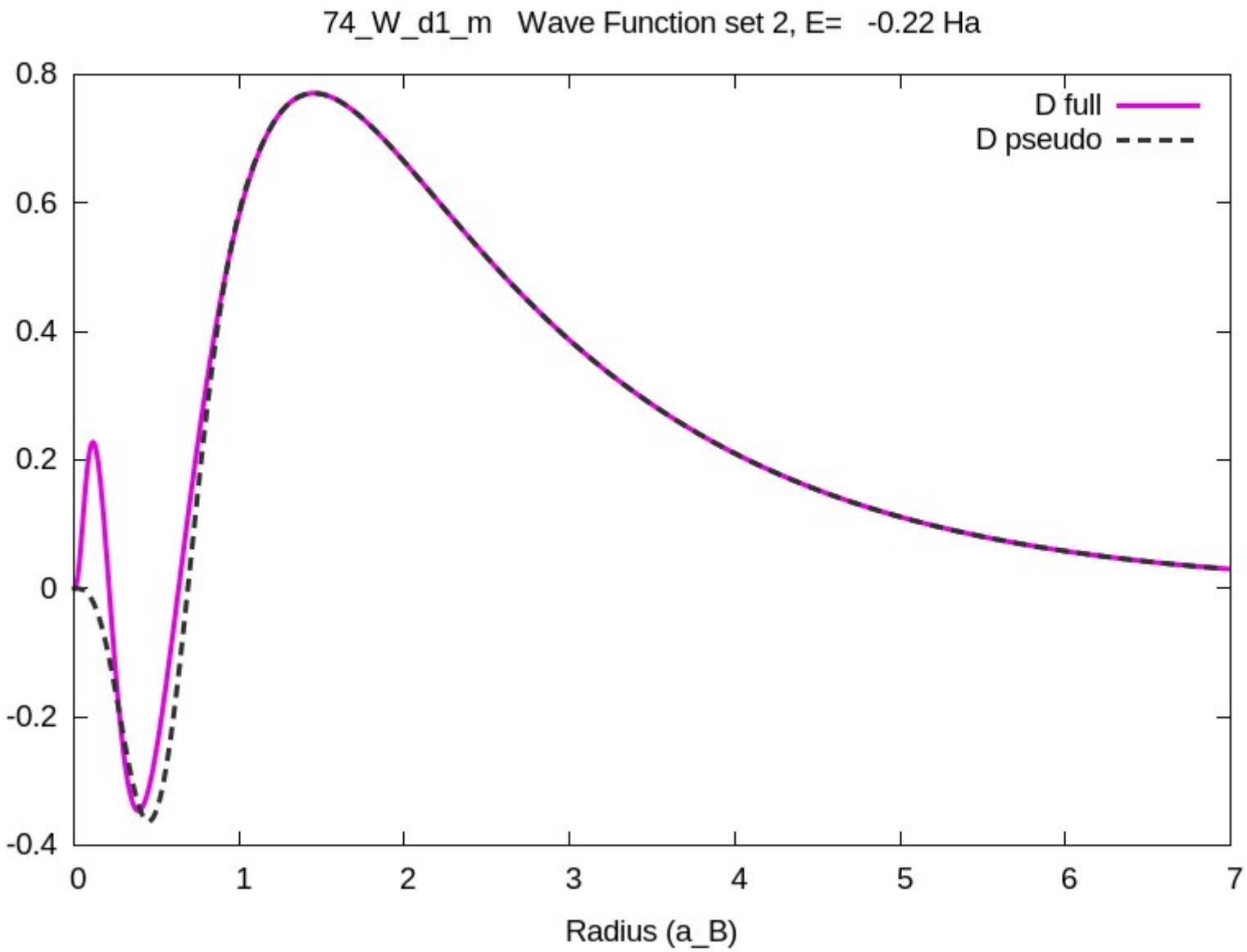

74_W_d1_m Wave Function set 1, E= -14.79 Ha
P full
P pseudo
2
1.5
1
0.5
0
-0.5
-1
0
1
2
3
4
5
6
7
Radius (a_B)

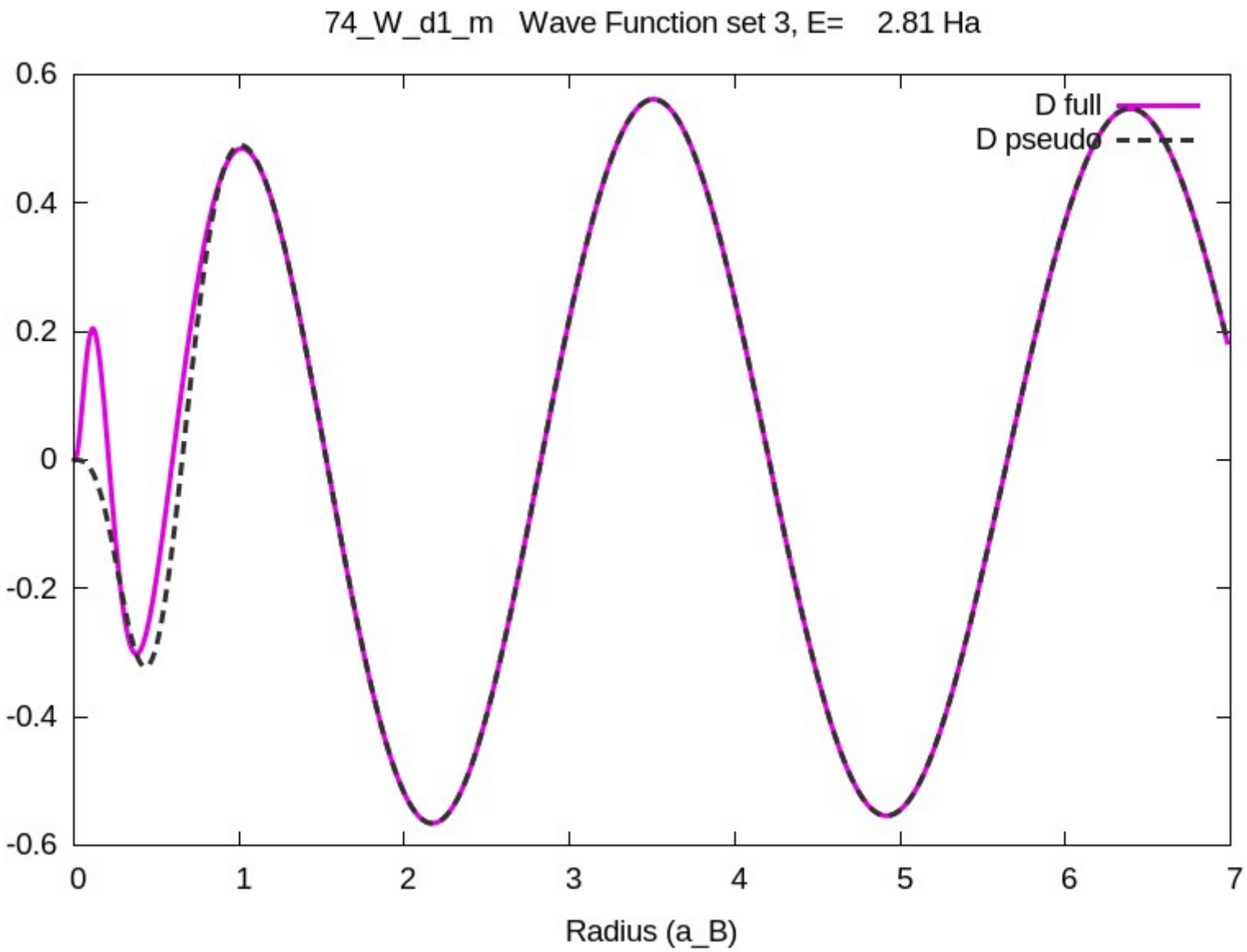

74_W_d1_m Wave Function set 2, E= -1.55 Ha
P full
P pseudo
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
1
2
3
4
5
6
7
Radius (a_B)

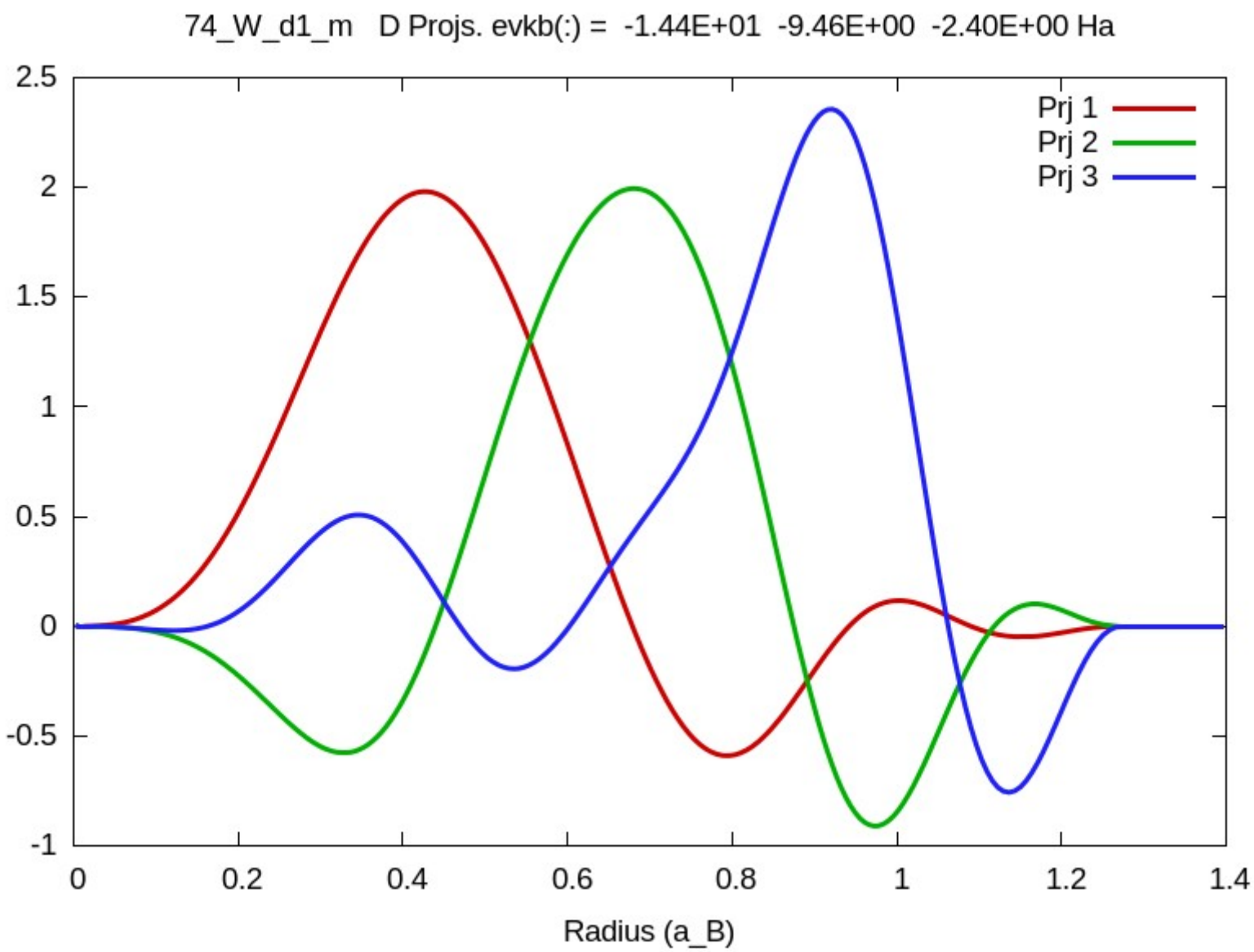

74_W_d1_m Wave Function set 3, E= 1.75 Ha
P full
P pseudo
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
1
2
3
4
5
6
7
Radius (a_B)

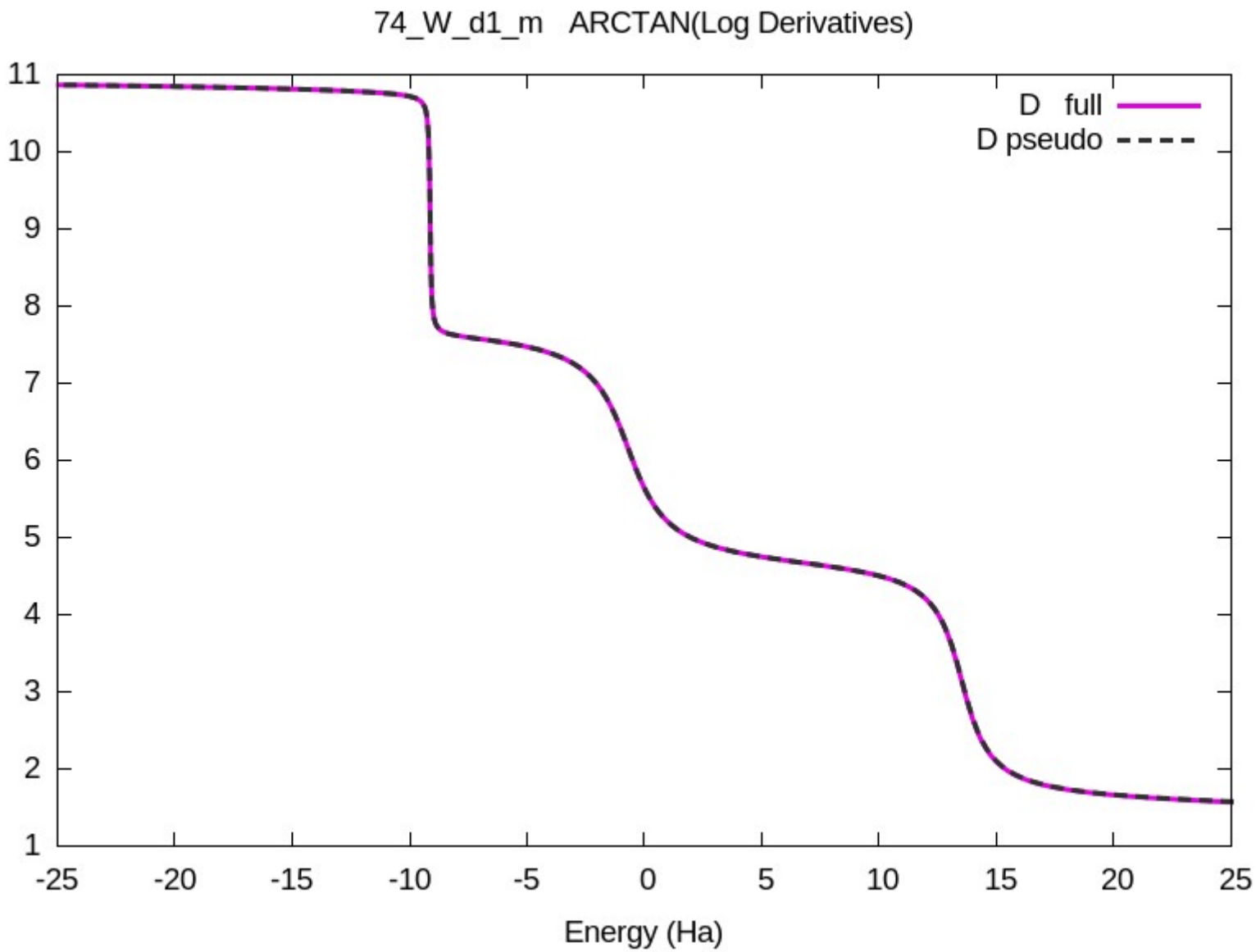

74_W_d1_m P Projs. evkb(:) = -8.57E+00 -2.82E+00 -1.11E+00 Ha
Prj 1
Prj 2
Prj 3
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
Radius (a_B)

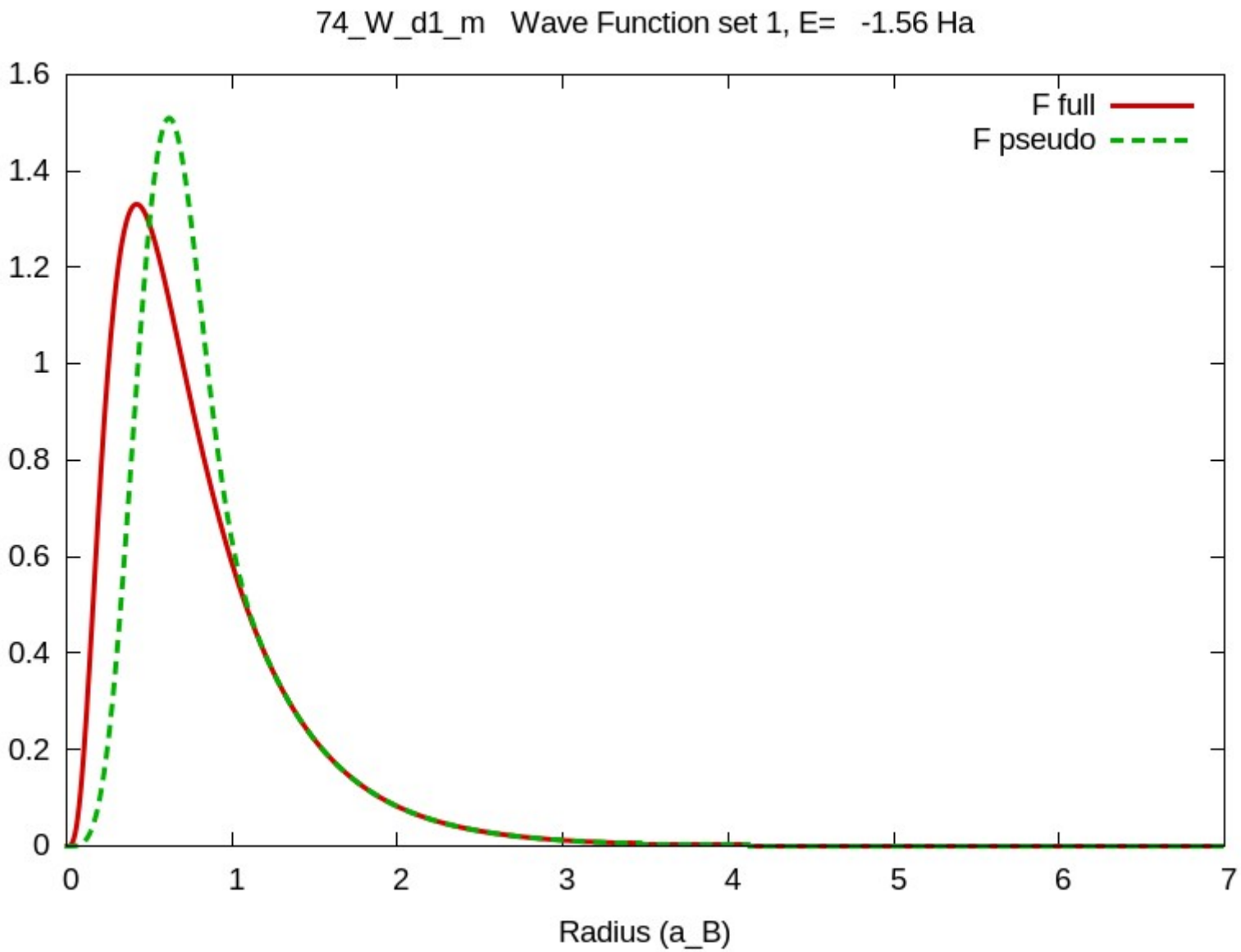

74_W_d1_m Wave Function set 1, E= -9.04 Ha
D full
D pseudo
2
1.5
1
0.5
0
-0.5
-1
0
1
2
3
4
5
6
7
Radius (a_B)

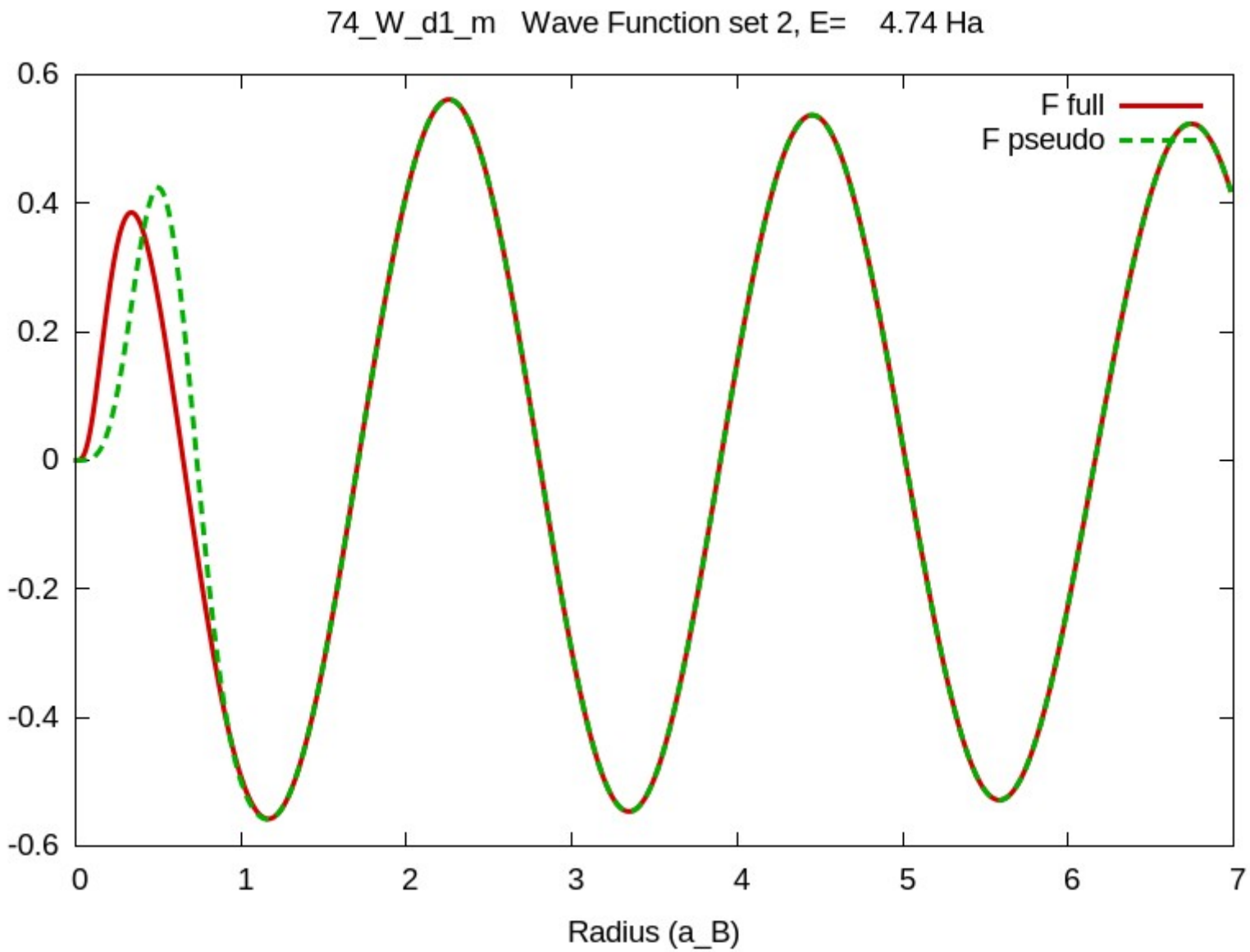

74_W_d1_m Wave Function set 2, E= -0.22 Ha
D full
D pseudo
0.8
0.6
0.4
0.2
0
-0.2
-0.4
0
1
2
3
4
5
6
7
Radius (a_B)

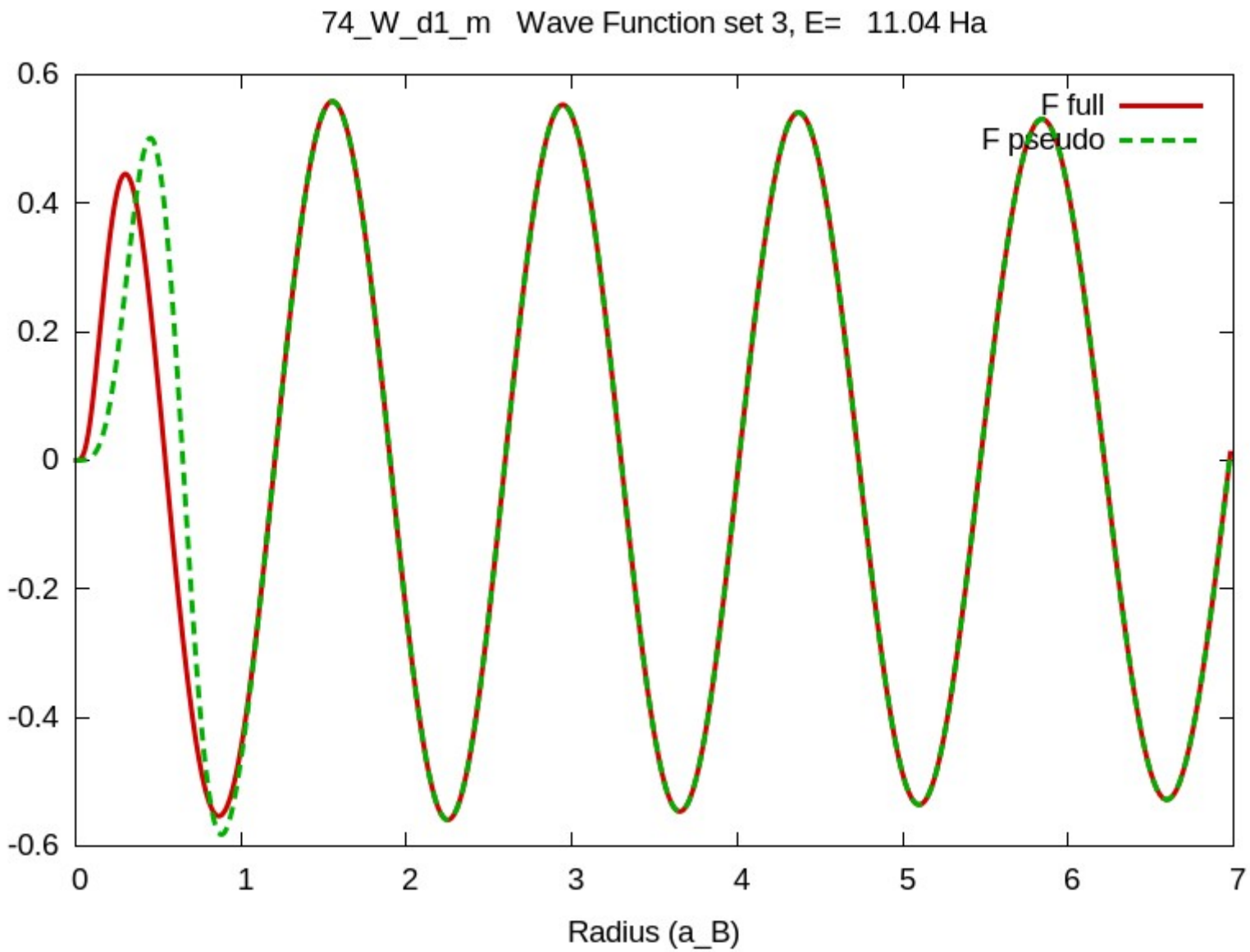

74_W_d1_m Wave Function set 3, E= 2.81 Ha
D full
D pseudo
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
1
2
3
4
5
6
7
Radius (a_B)

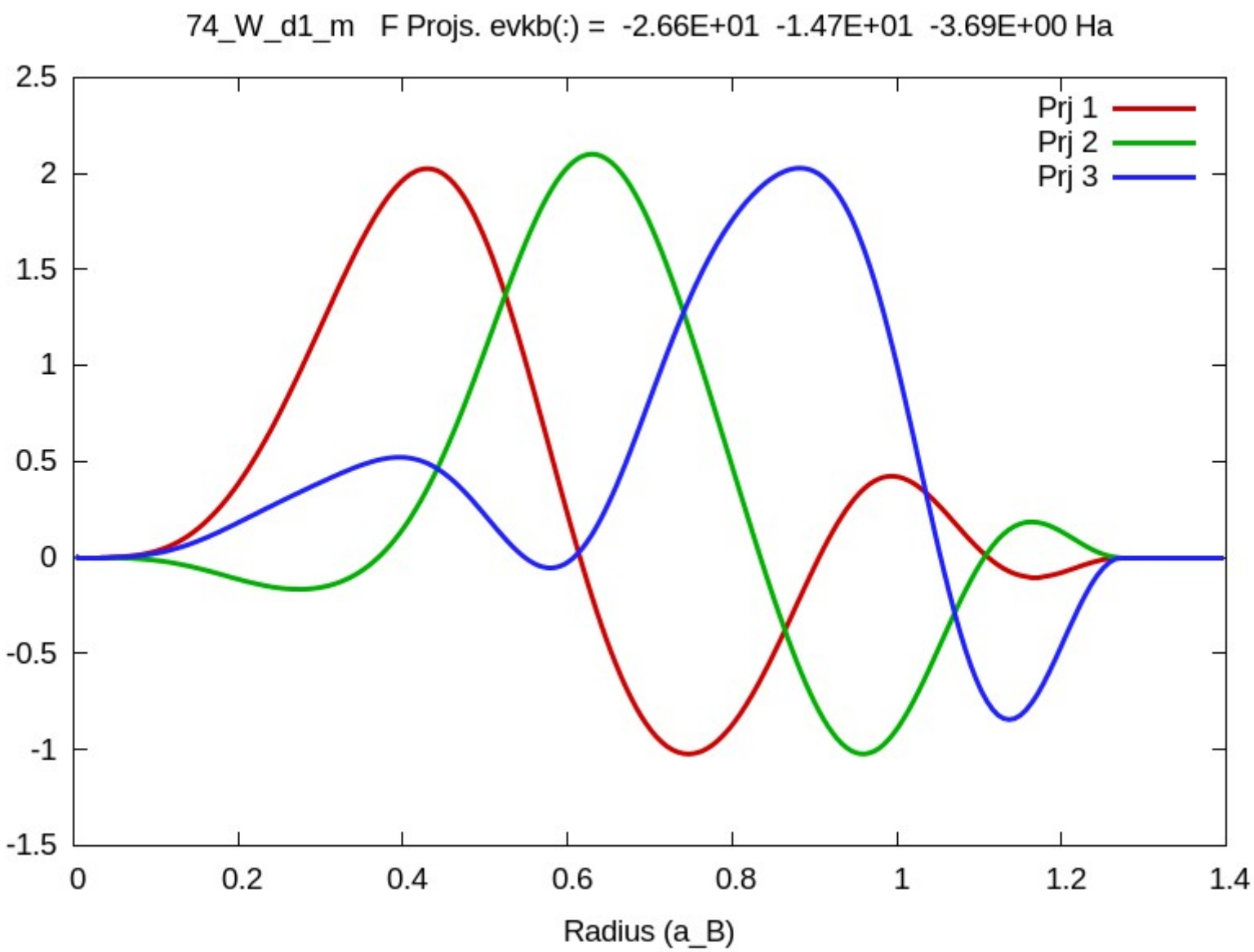

74_W_d1_m D Projs. evkb(:) = -1.44E+01 -9.46E+00 -2.40E+00 Ha
Prj 1
Prj 2
Prj 3
2.5
2
1.5
1
0.5
0
-0.5
-1
0
0.2
0.4
0.6
0.8
1
1.2
1.4
Radius (a_B)

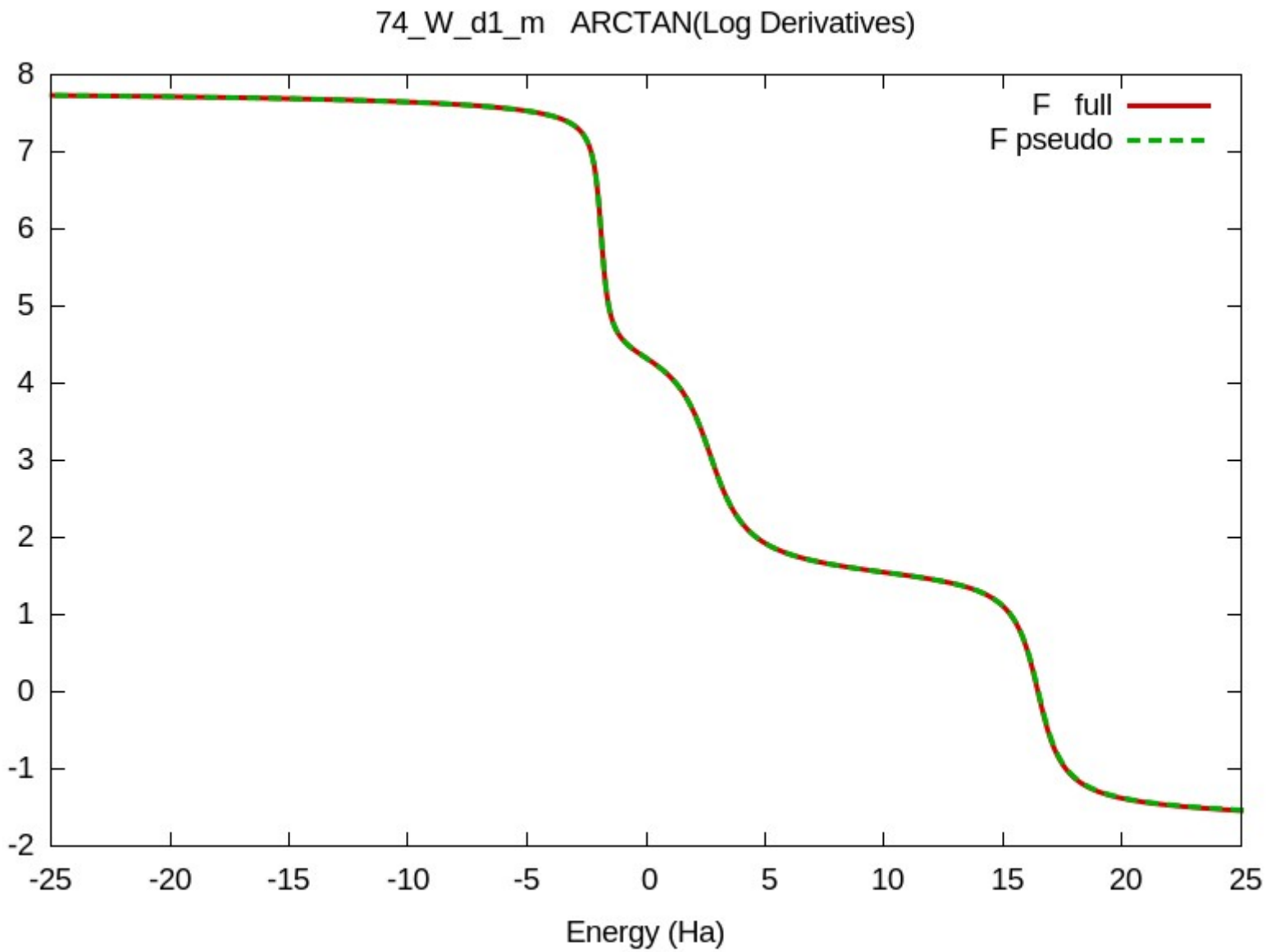

74_W_d1_m ARCTAN(Log Derivatives)
D full
D pseudo
11
10
9
8
7
6
5
4
3
2
1
-25
-20
-15
-10
-5
0
5
10
15
20
25
Energy (Ha)

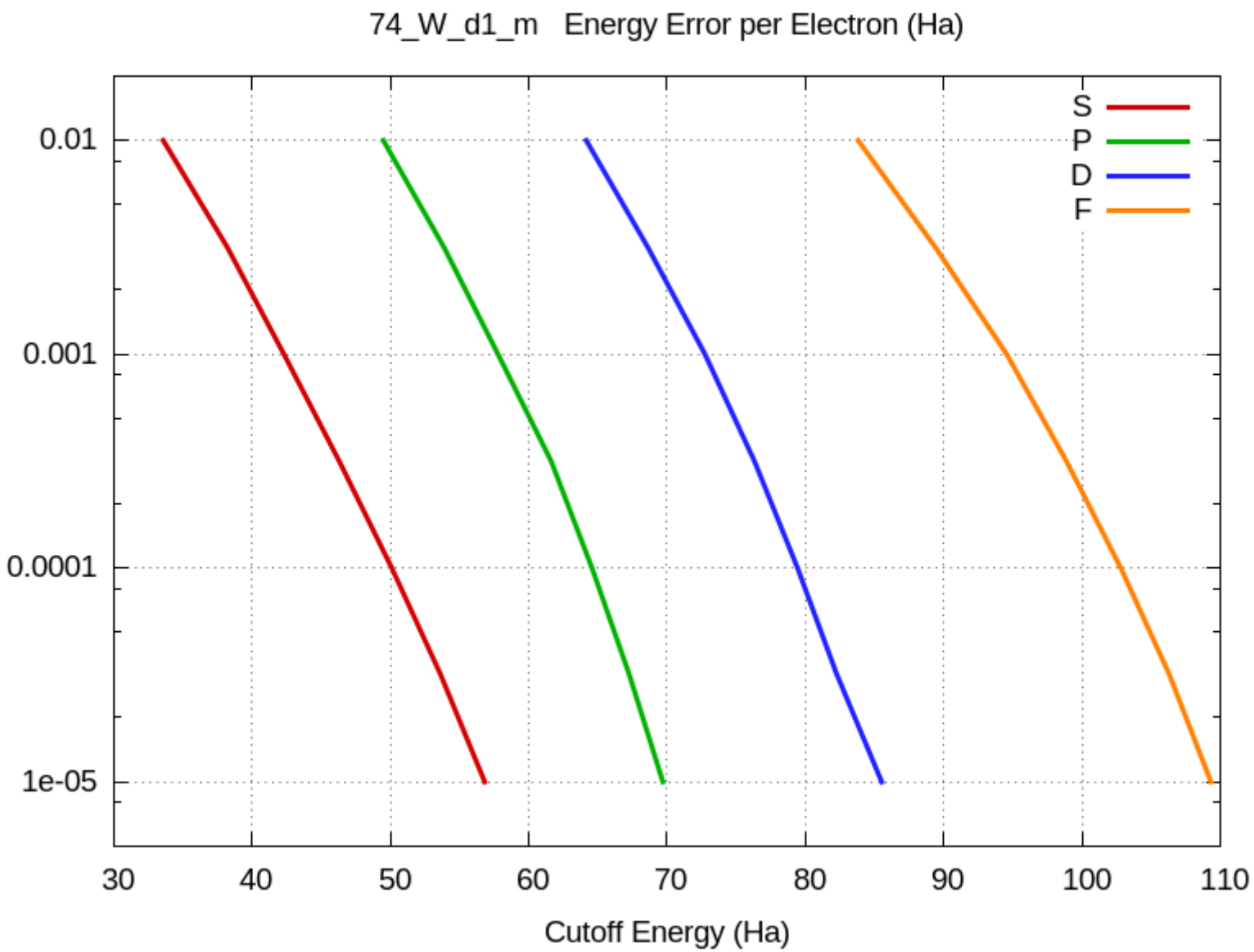

74_W_d1_m Wave Function set 1, E= -1.56 Ha
F full
F pseudo
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
1
2
3
4
5
6
7
Radius (a_B)


74_W_d1_m Wave Function set 2, E= 4.74 Ha
F full
F pseudo
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
1
2
3
4
5
6
7
Radius (a_B)


74_W_d1_m Wave Function set 3, E= 11.04 Ha
F full
F pseudo
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
1
2
3
4
5
6
7
Radius (a_B)


74_W_d1_m F Projs. evkb(:) = -2.66E+01 -1.47E+01 -3.69E+00 Ha
Prj 1
Prj 2
Prj 3
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
Radius (a_B)


74_W_d1_m ARCTAN(Log Derivatives)
F full
F pseudo
8
7
6
5
4
3
2
1
0
-1
-2
-25
-20
-15
-10
-5
0
5
10
15
20
25
Energy (Ha)


74_W_d1_m Energy Error per Electron (Ha)
0.01
0.001
0.0001
1e-05
30
40
50
60
70
80
90
100
110
Cutoff Energy (Ha)
S
P
D
F